

Initial Approaches to Treatment

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Abstract: This guideline provides an overview of principal findings and foundational principles for the optimal evaluation and management of workers. Selecting appropriate judicious tests and implementing optimal treatments with quality evidence of efficacy enhance patient recovery. It is important to emphasize functional recovery and provide the best treatment options starting with the first appointment to enhance the speed of recovery and prevent long-term disability that impairs quality of life. Key principles of recovery include avoiding or reducing substantially aggravating exposures, returning to work promptly and safely, encouraging active over passive treatments and exercise regimens, and patient education and active involvement. The reader is referred to ACOEM's other disorder-specific guidelines for more specific guidance.

Keywords: initial treatment, illness/injury, return to work, medications, exercise, physical methods, patient education, workplace interventions, functional recovery

The goal of treatment is to optimize the functional recovery of workers with work-related injuries, diseases, and/or disorders. This process includes regaining work-related and non-work-related quality of life, thereby eliminating or at least minimizing any residual disability. Providers may accelerate functional recovery through setting expectations, establishing measurable goals,¹⁻⁵ and selecting treatments that are proven to be effective in enhancing recovery, beginning with the initial, critically important clinic visit.

This is the second of the "Foundation Guidelines" of the American College of Occupational and Environmental Medicine's (ACOEM) Evidence-based Practice Guidelines that summarize principal findings and

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LEARNING OUTCOMES

- Identify the key principles of recovery for a work-related injury or illness.
- Describe initial treatments to administer to enhance recovery and prevent long-term disability.
- Describe the importance of addressing workplace interventions, including modified duty assignments and/or hazard control after the injury/illness.

foundational principles for the optimal evaluation and management of workers. This guideline focuses on the initial approaches to treatment. The first guideline addresses the General Approach to Initial Assessment and Documentation,⁶ whereas the other foundation guidelines include Independent Medical Examinations and Consultations,⁷ Work-relatedness,⁸ and Work Disability Prevention and Management.⁹ These guidelines underpin the many subsequent disorder-specific guidelines.¹⁰⁻²⁷

All of the ACOEM's Evidence-based Practice Guidelines are designed to provide health care providers with guidance for management of potentially work-related disorders among working-age adults. Although this guideline targets the initial treatment, the ACOEM guidelines also develop recommendations for the acute (up to 1-month duration), subacute (1- to 3-month duration), and chronic (more-than-3-month duration) clinical timeframes.

In developing this guideline, comprehensive search strategies were executed using five databases (PubMed, Scopus, CINAHL, Cochrane Library, and Google Scholar) without date limits following a standardized methodology. Searches were conducted for various evidence-based practice guidelines and medications (eg, acetaminophen, selective serotonin reuptake inhibitors, NSAIDs). Many searches were also performed in support of the other disorder-specific ACOEM guidelines and thus were not replicated for this guideline. Detailed search term write-ups of those searches are included in the respective disorder-specific guidelines.

A total of 1,134,216 articles were screened, and 1093 of their abstracts were reviewed and evaluated against specified inclusion and exclusion criteria. A total of 775 randomized controlled trials (RCTs) and 42 systematic reviews articles were included in this guideline. Evidence-based recommendations were developed and graded from "A" to

"C" in favor of and against the specific treatment, with A-level recommendations being supported by the highest-quality literature.²⁸ Expert consensus was employed for insufficient evidence (I) to develop consensus guidance.^{28,29} This guideline achieved 100% panel agreement for all of the developed guidance in this article.

Guidance was developed with sufficient detail to facilitate assessment of compliance (Institute of Medicine [IOM]) and auditing/monitoring (Appraisal of Guidelines for Research and Evaluation [AGREE II]).³⁰⁻³² Alternative options to manage disorders are provided when comparative trials are available.^{10,13,14,16-18,20,25,27} All AGREE II,³² Institute of Medicine,³³ AMSTAR,³⁴ and GRADE³¹ criteria were adhered to. In accordance with the Institute of Medicine's Trustworthy Guidelines, this guideline underwent external peer review, and detailed records of the peer-review processes are kept, including responses to external peer reviewers.³¹

The expert panel and research team that developed this guideline have complete editorial independence from the ACOEM and Reed Group, the publisher of the ACOEM's Guidelines, which have not influenced this guideline. The literature is continuously monitored and formally appraised for evidence that would materially affect this guidance. In contrast with disorder-specific guidelines where changes in evidence and standards for quality guidelines require more frequent updating, this guideline addresses relatively timeless principles and is planned to be updated should changes in evidence require it. All treatment recommendations are based on synthesis of the evidence in addition to expert consensus. These are recommendations for practitioners, and decisions to adopt a particular course of action must be made by trained practitioners and based on available resources and the particular circumstances presented by the individual patient.

OVERVIEW OF PRINCIPLES

This guideline provides an overview of strategies to optimize patient functional recovery in the early stages after injury in the absence of red-flag symptoms (ie, patient responses and findings that raise suspicion of serious underlying medical conditions, such as fracture, cancer, and infectious diseases). Red-flag symptoms should prompt evaluation for serious underlying disorder(s), with treatment then directed toward those disorder(s), for which the patient may need a referral to another provider for a thorough evaluation of a disorder not suspected to be work related. Red-flag symptoms are discussed further in the General Approach to Initial Assessment and Documentation guideline.⁶ In the absence of red-flag symptoms, which generally rules out the need for special studies, referral, or inpatient care, initial treatment of work-related injuries and illnesses is largely focused on managing patient expectations for recovery and generally utilizing noninvasive or minimally invasive treatments that have quality evidence supporting improved patient outcomes (see Table 1 for summary). The following principles apply to most work-related injuries and diseases:

- Musculoskeletal symptoms can be managed with activity modification, such as mitigation of or removal from substantially aggravating exposure(s). A short period of activities requiring modified use of the injured body part may be needed for severe injuries. Specific recommendations are also frequently helpful, including those addressing the following: employment, daily activities, home care, personal care, and recreational activities; short-term pharmacotherapy (usually oral but sometimes topical medication); a limited course of heat and/or cold therapy for acute but generally not chronic pain management (see disorder-specific guidelines for detailed information)^{10–27}; and non/occupational psychosocial factors, nutritional issues, and a disorder severity-appropriate course

- of physical and/or occupational therapy, especially for recovery in subacute to chronic cases.
- Eye injuries may require specific treatment such as removal of foreign bodies and rust rings. Blunt ocular trauma is also managed with evaluation for serious problems that require referral; otherwise, conservative management is employed. Activity modification and short-term pharmacotherapy (usually topical medication) are occasionally needed. Refer to the Eye Disorders guideline for detailed information.¹¹
- Respiratory symptoms can be managed with mitigation of or removal from substantially aggravating exposure(s). Specific recommendations may be provided regarding employment, daily activities, home care, personal care, and recreational activities, as well as short-term pharmacotherapy (usually inhaled but sometimes oral medication). Refer to the Occupational Asthma and Occupational Interstitial Lung Disease guidelines for detailed information.^{19,22}
- Most patients will experience decreased symptoms and improved physical functional abilities within days to weeks, although recovery time depends on the severity of the disorder and an extraordinarily complex array of other factors, such as workplace accommodations, psychosocial factors, prior injury history(ies), comorbidities, and degree of adherence to treatments. If recovery takes longer, the patient and provider should seek to identify other medical disorders, workplace exposures, avocational exposures, and psychosocial factors that may be preventing or delaying improvement or recovery.⁶
- For the disorders discussed in these guidelines, few patients need diagnostic tests to rule out a serious condition within the first several weeks. Aside from trauma and a few other exceptions, advanced diagnostic testing is not useful or cost-effective for most work-related musculoskeletal disorders in the first

- few days or weeks.^{25,27,35} In contrast, patients with respiratory disorders or eye injury often need diagnostic testing at the time of initial evaluation.
- Inactivity and/or immobilization should be limited due to concern for deconditioning, muscle weakness, bone loss, and development of incapacity or trend toward disability after relatively short periods of time.
- Progress in therapy or a directed home-exercise or activity program can be used as a means of increasing physical capacity and returning patients to function and work. Many patients benefit from instruction on specific exercises to address impairments and coaching in job tasks, under the direction of a physical therapist or occupational therapist.
- Safe return to work or work in a modified-duty capacity enhances the recovery of injured and ill workers (see Work Disability Prevention and Management guideline).⁹

PATIENT EDUCATION AND INVOLVEMENT

Patient education includes information on the diagnosis, prognosis, activity levels, activity modifications, work limitations, and treatment plan. Succinctness is generally necessary to avoid information overload and potential for missing the main points. The provider should address any identified fear-avoidant beliefs and the necessity of the patient to be involved in his or her recovery plan at the first visit.^{5,36,37} When patients are actively involved in decision-making, it may be easier to avoid inappropriate testing, streamline treatment, and hasten recovery. The need for discussion and information varies among patients and at various stages of care with some patients desiring more detailed information and discussion. It is helpful to discuss the uses and yields of diagnostic tests as well as the effectiveness and risks of proposed treatments in language that the patient will understand. Most studies of simple educational booklets

TABLE 1. Summary of General Recommendations on Initial Approaches to Treatment*

Treatment	Recommended	Optional	Not Recommended
Patient discussion, education, and involvement	Patient discussion		
Medication	Patient involvement NSAIDs Acetaminophen	Muscle relaxants Glucocorticosteroid injections Topical NSAIDs	Opioids, short 3- to 5-d course Other topical medications
Physical treatment methods	Early physical intervention with specific exercise prescriptions for a given specific disorder	Self-application of heat or cold Manipulation	Manipulation, with progressive or severe neurologic deficits

*See disorder-specific guidelines, as there are some disorders for which treatments may differ substantially from those in this generic table.^{10–27}

suggest that they are not effective when used without active treatments (see disorder-specific guidelines).^{10–27} Structured patient education alone may be less effective than other interventions but may be beneficial when combined with other interventions.^{38,39}

In the absence of symptoms or signs indicating a serious cause (ie, cancer, infection, or fracture), recently injured workers should be counseled to anticipate improvement in symptoms and functioning within a few days to weeks. Similarly, if the evaluation of a patient with subacute (1- to 3-month duration) or chronic (more than 3 months) symptoms has been reinvestigated and continues to not reveal a serious cause, the patient should be advised of the likelihood of a favorable functional outcome provided that he or she adheres to a functional restoration program. Self-care active methods such as exercises are, in most cases, critical for timely recovery. Active treatments such as specific, evidence-based exercise treatment plans tailored for a given disorder and physical conditioning generally offer considerably greater benefits than passive modalities.

Considering the patient's background and educational level, the following information should be provided:

- The natural history of the diagnosis.
- The generally favorable outlook for recovery supported by what is known about the injury or illness (assuming the disorder is acute and/or without long-term sequelae).
- The timeline for recovery, including goals and expectations of function.
- Testing and treatment options, with an explanation of the sensitivity, specificity, yield, risks, and benefits in lay terms and/or with print or audiovisual aids. (This is particularly necessary for major interventions; for modest options, such as over-the-counter medications, ice, and heat, detailed discussion is not needed.)
- Information to assuage fear-avoidant beliefs and other psychosocial factors. Early intervention to address psychosocial obstacles to recovery may be effective for improved outcomes. Such obstacles often include beliefs about the nature of the symptoms, prior injury history(ies), multiple comorbidities, whether harm occurs from use of the injured part, depression, job dissatisfaction, low perceived social support, financial concerns, employment security, and more.
- Safe return to work and normal function as primary expectations and collaborative goals. This includes counseling that patients who return early to full or modified work typically have superior short- and long-term outcomes.^{10,20,40–46}
- It may be helpful to summarize key points in the medical record, as information

overload is common with electronic medical records systems, especially given the dual nature of the medical and work-related issues of the visit. A general template of information could be created and distributed at the first visit to minimize redundancy with each new patient.

WORKPLACE ISSUES

Work accommodation and targeted provider communication with the workplace are usually quite helpful for effective early return to work. A provider taking an active role early in the return-to-work process achieves better results, especially by directly contacting the workplace. Work limitations or restrictions are best when they reflect the injured worker's current physical capacity and/or describe activities that should be avoided to reduce risk of harm to the worker or others. A job description from the employer, if available and with sufficiently detailed information regarding the physical and/or chemical demands of job functions and tasks, may help optimize the return to the workplace by providing information that may allow the provider to make more informed recommendations for specific work restrictions. It may also open the conversation for the potential of returning to or creating modified duty. It is often then helpful to discuss practical strategies for modifying the worksite to accommodate the worker and strategies to reduce the risk of recurrent injury, including addressing toxicological exposures, ergonomic factors, supervision, interpersonal factors, personal protective equipment, and task design. Only rarely do patients need to be removed from work entirely due to the extent of their injuries or risk to themselves or others. In these rare circumstances, it is helpful to frequently reevaluate the worker's capacity and provide a projected return-to-work date.

Satisfaction with a job, an employer's handling of a claim, and the medical care provided for a work-related injury may influence the patient's return-to-work prospects. Six to 12 months after an injury, patients reporting lower satisfaction with care are more likely to still be receiving lost-wage compensation. In the medical realm, higher levels of satisfaction have been associated with access to timely care, choice of provider, easy access to specialists, interpersonal behaviors during care, and having a worker-recovery orientation to care. Integrated case management improves patient satisfaction, resulting in more rapid return to work and fewer functional limitations at 6 months following injury.^{47,48}

MANAGING EXPECTATIONS

Total care management includes managing expectations and setting a trajectory toward rapid recovery by promptly instituting treatments that are most efficacious. Evidence

in support of the superiority of this approach among workers' compensation patients is relatively scarce,⁴⁹ but there is some evidence suggesting earlier institution and better compliance with evidence-based treatment guidelines results in less opioids use,^{50–52} lower costs incurred,^{49,51} and faster recovery with fewer lost workdays and/or less disability.^{50,51}

The provider sets the expectation for a patient's functional recovery at the initial visit and reinforces that expectation at subsequent visits. General information communicated to the patient about anticipated recovery may include population norms, results of quality treatment studies, and typical patient experiences. This overview should be further complemented by recovery estimates tailored to the individual patient.

Open discussion with the patient is helpful to understand the patient's knowledge, beliefs, and expectations about functional recovery. It is important to address misconceptions about the causes and meaning of symptoms and discuss recovery and preventive measures. Patients may believe that their symptoms signal a serious structural injury and that they will suffer further damage, be permanently disabled, or require surgery if they remain physically active. Addressing these potential misconceptions and fears can facilitate functional recovery. When medications, injections, or surgery are indicated, patients may require additional discussion or information to allay fears.

A high level of catastrophizing or kinesiophobia may increase the likelihood of chronic back pain and future disability. There are several screening tools available to assess which patients may have a greater risk of disability and may benefit from early targeted interventions addressing the nonmedical issues impacting the injury. Many offices have preliminary screens such as ACT-UP, depicted in Table 2, or other resources.

Inadequate assessment of an injury or of ability to return to work can negatively impact the worker's recovery as well. The provider should avoid catastrophizing the situation or promoting a patient's fear avoidance behavior(s). Motivational interviewing or similar may be an effective means of handling what can be a difficult discussion for many providers.⁵³

PATIENT COMFORT

Relief of pain is often the injured worker's major concern. Patients usually correlate their degree of discomfort with injury severity, which may lead to reluctance to participate in potentially therapeutic activities. Therefore, the patient may benefit from learning that his or her degree of discomfort may not correlate with the extent of injury for many musculoskeletal disorders. Patient discomfort may be alleviated with education and training to address all of (as appropriate):

TABLE 2. ACT-UP Preliminary Screening Questions

Activities	How is your pain affecting your life (ie, sleep, appetite, physical activities, and relationships)?
Coping	How do you deal/cope with your pain (what makes it better/worse)?
Think	Do you think your pain will ever get better?
Upset	Have you been feeling worried (anxious)/depressed (down, blue)?
People	How do people respond when you have pain?

- Specific activity prescriptions;
- Activity modification(s);
- Judicious activity limitations (which should be conveyed as applying to work and home situations);
- Physical methods (self-treatment and provided by a healthcare worker);
- Medication(s);
- Counseling about the nature of the injury to address concerns and reduce anxiety; and
- Emphasis on recovery of function.

If a patient does not recover as quickly as expected, it is helpful to seek and address the reasons for delay. Patients who do not improve within a few days with appropriate medical treatment and consideration of workplace exacerbating factors (eg, those with eye symptoms, many with occupational asthma, most with mild low back pain) or weeks (eg, those with moderate to severe musculoskeletal symptoms) may need additional evaluation to identify physical factors or medical causes for the delay, treatment noncompliance, a change in treatment strategy, or additional intervention to address psychological or social contributors to the delayed recovery. Besides providing the patient with a realistic set of expectations, one must manage the expectations of the family, employer, insurance carrier, and perhaps a union or lawyer. Written and verbal contact can keep these parties educated and informed.

RELATIVE REST, IMMOBILIZATION, AND ACTIVITY

Elimination of exposure is important in certain discrete disorders (eg, occupational asthma [allergic] secondary to a specific chemical or allergic contact dermatitis). For most musculoskeletal injuries, however, restrictions of activity and immobilization result in deconditioning and bone loss within a matter of days and often delay recovery. Bone or muscle lost from restriction of activity or immobilization cannot be restored without undertaking exercises, resumption of activity levels, or a formal reconditioning program. Aching, stiffness, and pain will often occur if muscles and joints are not used. Mobilization of painful areas often helps reduce pain. Depending on the disorder in question, guided aerobic and specific activities may improve comfort both acutely and as recovery progresses; target heart rate and/or progressive benchmarks of exercises are

advisable. Early mobilization has not been associated with increased complications, deformity, or increased residual symptoms. Reported benefits of mobilization have included earlier return to work, decreased pain, swelling, and stiffness, and improved range of motion.^{20,25,27}

GENERAL PRINCIPLES OF TREATMENT

Discussion and recommendations for each disorder are provided in disorder-specific guidelines.^{10–27,54,55} However, general principles of treatment apply broadly across all guidelines. These are comprised of “treatment” addressing the workplace, such as workplace interventions, including modified duty assignments and/or hazard control such as in the hierarchy of hazard controls: elimination or substitution, engineering controls, administrative controls, and personal protective equipment. Aspects of treatment also focus on the individual. These include assessment of whether prior medical disorders may impact or be impacted by the current disorder; addressing contributing social, psychosocial, and wellness factors; and referral to workplace wellness programs,⁵⁶ physical methods including disorder-specific exercises, the numbers of medications to prescribe, generic compared to trade medications, compounded medications, numbers of medications, numbers of treatments and modalities, length of treatments, measures of function, and telemedicine indications. These are reviewed hereinafter.

Workplace Job Interventions

Workplace interventions may reduce or eliminate the period of absence among workers with respiratory and dermatological disorders related to toxicological exposures. The effectiveness of workplace interventions on work disability ranges from modest to large. Workplace interventions may reduce time to return to work and improve pain and functional status in workers with musculoskeletal disorders.⁵⁷ Limited evidence indicates that material handling education and training without either redesign or assistive devices do not prevent back pain and pain-related disability or reduce sick leave when compared to no intervention or alternative interventions. However, the absence of quality evidence does not mean that there is a lack of efficacy, and further studies are needed.

Lifestyle interventions also have some evidence of efficacy.⁵⁶

Workplace hazard control, mitigation or medical removal from exposure, and use of personal protective equipment may all be part of a treatment plan. It is important, especially for occupational diseases, to identify the exposure source and institute hazard control measures to prevent further exposure. Typically, the patient is a good source of information as to how the injury occurred and sometimes a good source for recommendations for the solution. This can be the basis for discussion of recommendations with the employer regarding restrictions or hazard elimination or control. A site visit by a trained professional (eg, occupational medicine physician, safety professional, ergonomist, industrial hygienist, occupational therapist, physical therapist, or occupational health nurse) may be useful if these resources are available.

If the source of exposure cannot be identified or controlled, or if the illness is such that immediate elimination of any exposure is necessary, medical removal from the workplace is an option. Personal protective equipment may reduce or prevent exposures but should ideally be implemented only as an adjunct to engineering and administrative hazard controls. Follow-up after return to work is important to ensure that the worker and employer are complying with restrictions and to determine whether restrictions are sufficient, excessive, or unable to be accommodated.

Medications

Medications are selectively recommended for the treatment of pain (Evidences A–C). Evidence suggests that age and the number of medications are associated with adverse drug reactions^{58–62} with large data sets suggesting that the risks of adverse effects with more medications are exponential.^{38,58,60–63} The number of prescribers is also a reported risk for adverse drug reactions.⁶⁴ Limiting numbers of medications and discontinuing ineffective medications are not invasive, result in reductions in risk, and lower costs and thus are recommended (Evidence I).

Generally, a trial of one medication for a specific goal (eg, pain reduction) should be prescribed at a time and evaluated for efficacy especially regarding objective and functional improvements in preference to subjective measures (see Table 3). Generic medications are nearly always thought to have identical components and thus comparable efficacy as trade-name medications.^{65,66} Evidence of efficacy for a medication in the ACOEM Guidelines and in general medical practice is developed based on the drug class of the pharmaceutical. In the absence of evidence to the contrary, there is a reasonable presumption that drugs within the same class have the same degree of efficacy, Evidence (I). When there are cost differences, the lower-cost

option is preferable, Evidence (I). Individual generic medications (or trade medication[s] if necessary) are Evidence (I) as preferential to compounded medications, which are generally more expensive and nearly universally without quality evidence of efficacy. Medications should be selected with quality evidence of efficacy for that specific disorder.^{10–27} In select cases and especially for acute evaluations, two medications and infrequently three may be reasonable (eg, a nonsteroidal anti-inflammatory drug [NSAID] plus a muscle relaxant at night to assist with sleep); the main exceptions are the immediate postsurgical time and severe accidents. However, in both of those examples, it is generally preferential to use lower doses of potentially impairing medications during the day and increase physical activity at the earliest opportunities. Quality evidence in support of combination(s) of medications and/or other treatments is otherwise generally quite rare in evidence-based medicine (see disorder-specific guidelines for exceptions).^{10–27} Medication use should cease when there is end of healing, noncompliance, and/or plateau. Resumption of medication may be reasonable if there is demonstrated need after cessation.

Oral Pharmaceuticals

Oral pharmaceuticals are a first-line palliative method to treat pain and facilitate increased activity. Over-the-counter (OTC) analgesics typically provide sufficient pain relief for most patients with acute work-related symptoms. Time-limited prescription NSAIDs are a reasonable first-line option for some patients. Nearly all quality trials utilized scheduled administrations of medication rather than “as needed” prescriptions; thus, utility of unscheduled “PRN” doses are of uncertain benefit. If treatment response to

oral analgesics is inadequate (ie, symptoms and activity limitations continue without significant improvement), the provider should reevaluate the diagnosis, assess whether physical methods (exercise, modalities, etc) are either in need of adjustments or implementation, ascertain if the medication is being used as directed, and assess workplace and other physical activity levels. If the exercises prescribed, over-the-counter medication use, and activity levels are appropriate, the provider should consider treatment with prescribed pharmaceuticals and/or additional physical methods. Consideration of comorbid conditions, adverse effects, cost, efficacy, and patient preferences should guide the treatment recommendations. The provider should discuss the efficacy of medication for the disorder, any adverse effects, and any other relevant information to ensure proper use and to manage expectations. Also important is to be aware of the therapeutic dose for the disorder under treatment (eg, gabapentin for neuropathic pain) and individualizing to patient age and body mass. Ongoing use of medication, as with all other interventions, should be guided by objective evidence of functional improvement and should be coupled with an active treatment regimen.

Nonsteroidal Anti-inflammatory Medications and Acetaminophen

Quality evidence indicates that NSAIDs, including aspirin and ibuprofen, are more effective than acetaminophen in RCTs for treatment of many musculoskeletal disorders (see Low Back Disorders and Knee Disorders guidelines).^{18,67} For patients with milder pain or medical contraindications for NSAID use, acetaminophen is an option for pain relief. Acetaminophen can be used in combination

with NSAIDs or other pharmacologic methods.

There are unresolved questions as to whether NSAIDs interfere with the fracture healing process. However, the evidence supporting a risk of NSAIDs as causing nonunions is very low quality (ie, not RCTs). It is also concerning that the odds ratio for nonunion for opioids use is higher than for NSAIDs.⁶⁸ Thus, there is a strong reason to believe that these relationships are either partially or totally confounded, whereby those using the medications are also the ones with the nonunions.⁶⁹

NSAIDs are associated with potential adverse gastrointestinal, cardiovascular, renal, hepatic, and allergic effects. Gastrointestinal adverse effects are particularly problematic for the elderly, rather than healthy working-age populations, and may exclude some occupational patients from NSAID use. Nonetheless, COX-2 inhibitors have been found to have significantly reduced the risk of major GI events, especially in individuals at risk of NSAID-related adverse effects.⁷⁰ Similarly, renal adverse effects may be problematic, but they are most typical in the elderly or those with systemic diseases such as diabetic nephropathy. A large-scale study has evaluated the safety profile of celecoxib, naproxen, and ibuprofen and found that the risk of renal and gastrointestinal events was lower with celecoxib.⁷¹ Providers should weigh the risks (adverse effects and potential drug interactions) and benefits of NSAID use for individual patients (see Hip and Groin Disorders guideline for discussion of adverse NSAID effects and role of cytoprotective agents).¹⁷ These issues tend to be considerably less problematic among employed patients due to their generally younger age, better overall health, and shorter treatment courses than the elderly and others considered at high risk to develop NSAID complications. If any NSAID is to be used for a prolonged period, such as more than 4 weeks, consider the efficacy prior to continuation and review potential adverse reactions and monitoring recommendations. For example, diclofenac prescriptions are recommended to be assessed with hepatic enzymes aspartate aminotransferase (AST)/alanine transaminase (ALT) at 4 to 8 weeks of treatment initiation, and then if long-term, a complete blood cell count should be performed periodically. Cyclooxygenase inhibitors, although costlier than the NSAIDs, may be an appropriate choice for some patients.

A practice guideline provides an algorithm for prescription of NSAIDs and COX-2 inhibitors and proton pump inhibitors (PPIs) based upon cardiovascular and gastrointestinal risk factors.⁷² The use of PPIs should consider not only their benefits but also their potential harms.⁷³

TABLE 3. Examples of Objective Function-Based Goals and Secondary Goals to Track During Treatment

Primary Functional Goals

1. Return to work from nonworking status
2. Return to full duty work from modified working status
3. Advancement of activity, especially observed in therapy
 - a. Increased weight lifted
 - b. Increased numbers of repetitions and/or sets
 - c. Increased distance walked
 - d. increased pacing, work simulated tasks, cardiovascular capacity, and endurance

Secondary Functional Goals

1. Resumptions of activities of daily living (eg, clothing, bathing, showering)
2. (Re)establishment of sleep hygiene
3. Resumption of household chores
4. Resumption of sports
5. Validated functional instruments*

*Generally, functional instruments are subjective and lack objective measures.

Opioids

Opioids appear to be no more effective than safer analgesics for the management of most musculoskeletal symptoms but are recognized as key factors underlying the greatest epidemic of healthcare-related fatalities over the past century (see Opioids guideline).²⁴ Data on efficacy compared with placebo show only an approximately 1/11 pain rating reduction for acute pain⁷⁴ and 0.69/11 for chronic pain.⁷⁵ Opioids should be used only if needed for severe pain and then limited to low doses (20–50 morphine mg equivalents), nocturnal rather than round-the-clock use, and rarely beyond a short time frame of 3 to 5 days. Providers should counsel patients on the risks and adverse effects associated with opioid use, including death, motor vehicle crashes, falls, drowsiness, clouded judgment, memory loss, greater risk of disability, constipation, and potential for misuse or dependence (see Opioids guideline).²⁴ Adverse effects have been reported in up to 80% of patients taking opioid medications.

Injections

Injections of glucocorticosteroids, local anesthetics, or both should generally be reserved for patients who do not improve with more conservative therapies, although there are rare exceptions where the initial intervention may best be an injection, including trigger digit and de Quervain's (see Hand, Wrist, and Forearm Disorders guideline).²⁵ There are no quality comparative trials to guide the selection of a specific glucocorticoid medication for therapeutic injections and few quality data comparing doses (see disorder-specific guidelines).^{10–27} The medication used and frequency of injection should be guided by the goal of the injection (ie, diagnostic or therapeutic), the underlying musculoskeletal diagnosis, and clinical experience. Use of ultrasound to guide injection has not been clearly demonstrated to result in superior outcomes; thus, there is no requirement for its use, although it may be essential for those who learned injections only using ultrasound guidance.

Alternative Medicine

A few complementary/alternative agents such as ginger and turmeric have some evidence suggesting possible benefit for a few select disorders (see disorder-specific guidelines).^{10–27} Although there may possibly be a better safety profile than NSAIDs in the elderly, at higher doses of these agents, heightened bleeding risks, oxalate stones, and hepatic toxicity have been reported.

Physical Methods

Treatment modalities utilized by physical therapists, occupational therapists, chiropractors,

physicians, and other health care practitioners are sometimes broadly categorized as physical methods. These treatments frequently include specific exercises, acupuncture, thermal modalities (eg, moist heat, ultrasound), and manual therapies (eg, manipulation, massage, muscle energy techniques, or proprioceptive neuromuscular facilitation). Therapists often apply specific modalities and therapeutic exercises based theoretically on the stages of healing. A typical therapy protocol progresses sequentially through the following theoretical phases that may overlap: pain control, restoration of range of motion, restoration of strength, neuromuscular retraining, restoration of cardiovascular capacity, endurance, and return to full activity. Chiropractors usually emphasize manipulation, but many also frequently utilize other physical methods including exercise and other modalities.

Some providers relinquish therapy to the physical and/or occupational therapist and are not sufficiently engaged in the process. However, physical and occupational therapy should ideally be a collaborative approach. Active treatments, especially exercise, have the best evidence for efficacy and differ markedly by the type of the disorder (see disorder-specific guidelines).^{10–27} and evidence also suggests that function-based approaches are more successful.⁷⁶ The provider should ascertain that the physical and/or occupational therapy providers' time is spent predominantly in active treatment, engaging the patient and providing an appropriate home exercise program with components shown to be efficacious for the specific disorder. Patients should subsequently be able to convey and demonstrate to the provider in follow-up visits the treatments they are undergoing and the exercises they are performing. Most passive modalities have either been shown to be unsuccessful or to result in low-magnitude benefits. Judicious use of passive modalities may have a role in acute injury care if they offer sufficient symptom relief to increase participation in active therapy but should be time limited.

During the acute phase, providers may recommend application of heat and cold (aka, cryotherapies) for temporary amelioration of symptoms to facilitate mobilization and graded exercise. The use of cold in the management of acute soft tissue injuries is widely practiced, but there is insufficient evidence to prove that it advances clinical outcomes in the management of soft tissue injuries. For treatment of acute low back pain, there is more quality evidence of efficacy for application of heat than for cryotherapies.⁶⁷ Heat and cold treatments may be most effective when applied several times a day as tolerated, yet there are no quality studies comparing numbers of such treatments to determine an optimal approach.

Moderate- to high-quality evidence supports the therapeutic effectiveness of

manual therapies, exercise, and acupuncture in the management of some categories of musculoskeletal pain; however, magnitudes of benefits for passive modalities, if effective at all, are modest. Manipulative therapy may expedite the recovery of patients with acute low back pain of less than 4 weeks' duration, although it is noteworthy that attempted sham-controlled and higher-quality randomized trials conflict. The evidence is less strong regarding the benefit of spinal manipulation for patients with subacute or chronic low back pain (see Low Back Disorders guideline).⁶⁷

The indication for ongoing use of manipulation, mobilization, or modalities should be guided by objective evidence of ongoing, incremental functional improvements and should be coupled with an active treatment regimen. Lack of incremental improvement, particularly objective data (eg, grip strength, distance walked, range of motion, sets and numbers of specific strengthening exercises), is a strong indication for either substantive changes in the therapeutic approach or, more commonly, cessation of treatment(s). The value of therapy is believed to considerably increase when there is vigilant attention to the process of rehabilitation on the part of all providers. This also necessitates sufficient communication between the patient's providers to ensure that the information given to the patient is consistent and without conflict. Communications should generally include development of treatment goals, essential modalities, and an emphasis on training in home-based treatments. Communication with the therapist may also be of assistance to monitor a patient's physical status, psychosocial issues raised with the therapist, motivation, compliance with treatment and home exercise (or respiratory) recommendations, and functional progress. Therapists and providers should periodically have the patient demonstrate exercises to verify correct technique. Providers may then be in an excellent position to firmly reinforce the directions given to the patient and provide substantial encouragement to aid in return to work.

Patient beliefs regarding the potential benefits of physical modalities may influence the effectiveness of these modalities. Providers should counsel and educate patients about the nature and anticipated benefits of modalities to help them understand their role and to enhance outcomes.

Physical Method Prescriptions

Generally, a trial of one treatment (eg, specific exercises, iontophoresis, manipulation, acupuncture, ultrasound) is recommended (Evidence I) to be provided at a time and evaluated for efficacy especially using measures of functional gain (see Table 3). Utilizing a functional test/retest method within a treatment session helps confirm effective treatment and allows quick progression to the next functional level

within the same treatment session. In select cases and especially for acute evaluations, or occasionally when there is a change of provider with a need to institute efficacious treatments from a non-evidence-based regimen, two treatments may be reasonable. Quality evidence in support of combination(s) of medications and/or other treatments is generally quite rare in evidence-based medicine (see disorder-specific guidelines for exceptions).^{10–27} Patients should be provided a treatment that has quality evidence of efficacy. Ineffective treatment(s) should be discontinued prior to provision of alternate treatment(s). Prescriptions typically of approximately four to six appointments should be provided, and then functional gain should be assessed. There is no specific limit to numbers of appointments or treatments. Additional sets of four to six appointments are reasonable provided there is further, incremental functional gain. Additional appointments should cease when there is end of healing, noncompliance, and/or plateau. Observations should emphasize objective measures of functional gain in preference to subjective measures and/or subjective functional instruments (see Table 3). Failure to track improvements is believed to result in needless suffering, delayed recovery and delayed return to work. Limiting numbers of treatments and discontinuing ineffective treatments result in reductions in risk and lower costs and thus are also recommended (Evidence I).

Distance-Based Services (Telehealth)

It is always preferable to take an in-person history and perform a physical examination, and those are especially desirable for the first visit, for any consultation, and/or when the patient's trajectory is not following an expected rate of recovery. Although there is only limited evidence of efficacy of distance-based services (ie, telemedicine, telehealth),^{77–83} there are circumstances when the provision of distance-based health may be preferable, if not necessary. These include long distances relative to the value obtained during an office visit. Thus, telehealth is selectively recommended (Evidence I).

SUMMARY

Optimal management of the patient's initial treatment encounter sets a trajectory that can then facilitate functional recovery that includes reducing or eliminating symptoms. Emphasizing functional recovery and providing the best treatment options starting with the first appointment enhance the speed of recovery as well as prevent long-term disability that impairs quality of life. The provider should set patient expectations for regaining quality of life and quality of work life ideally from the time of initial injury. Clear communication and coordination of care with the patient and employer are critical and are also thought to help prevent disability.

Collaborative interventions and integrated care are often helpful, especially for complex cases and chronic pain. Selecting appropriate judicious tests and implementing optimal treatments with quality evidence of efficacy further enhance recovery. A few tenets of recovery include avoiding or reducing substantially aggravating exposures, returning to work promptly and safely, and encouraging active over passive treatments and exercise regimens. Patient education and active involvement are valuable cornerstones. Thoughtful integration of knowledge from the entirety of the disorder-specific treatment guidelines is optimal.

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REFERENCES

- Flood AB, Lorence DP, Ding J, McPherson K, Black NA. The role of expectations in patients' reports of post-operative outcomes and improvement following therapy. *Med Care* 1993;31:1043–1056.
- Main CJ, Buchbinder R, Porcheret M, Foster N. Addressing patient beliefs and expectations in the consultation. *Best Pract Res Clin Rheumatol* 2010;24:219–225.
- Bovend'Eerdt TJ, Botell RE, Wade DT. Writing SMART rehabilitation goals and achieving goal attainment scaling: a practical guide. *Clin Rehabil* 2009;23:352–361.
- Mahomed NN, Liang MH, Cook EF, et al. The importance of patient expectations in predicting functional outcomes after total joint arthroplasty. *J Rheumatol* 2002;29:1273–1279.
- Lorig KR, Holman H. Self-management education: history, definition, outcomes, and mechanisms. *Ann Behav Med* 2003;26:1–7.
- Klees JE, Milek D, Hegmann KT, ACOEM Evidence-based Practice Initial Assessment Panel. General approach to initial assessment and documentation [published online October 27, 2025]. *J Occup Environ Med*. doi:10.1097/JOM.00000000000003591.
- Stewart-Patterson CH, Els C, Zipper R, Hegmann KT. *Independent Medical Examinations and Consultations*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2018: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Greaves WW, Das R, McKenzie JG, Sinclair DC 2nd, Hegmann KT. Work-relatedness. *J Occup Environ Med* 2018;60:e640–e646.
- Kertay L, Caruso GM, Baker NA, et al. Work disability prevention and management. *J Occup Environ Med* 2025;67:e267–e280.
- Hegmann KT, Feinberg SD, Aronoff GM, et al. *Chronic Pain*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2024: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Blais BB, Anshel JR, Glass LS, Saito K. *Eye Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2017: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Parks P, Conidi FX, et al. *Traumatic Brain Injury*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2025: Available at: www.mdguidelines.com. Accessed October 8, 2025.
- Hegmann KT, Hughes MH, Haas NS, et al. *Ankle and Foot Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2018: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Hoffman HE, Belcourt RM, et al. *Elbow Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2018: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Bruns D, Warren PA, et al. *Posttraumatic Stress Disorder and Acute Stress Disorder*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2025: Available at: www.mdguidelines.com. Accessed October 3, 2025.
- Hegmann KT, Travis R, Andersson GBJ, et al. *Cervical and Thoracic Spine*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2018: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Guanche CA, Famath AF, et al. *Hip and Groin Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2019: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Biggs JJ, Hughes MH, et al. *Knee Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2019: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Biggs JJ, Hughes MH, et al. *Interstitial Lung Disease*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2019: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Travis R, Andersson GBJ, et al. *Low Back Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2020: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Bruns D, Warren PA, et al. *Depressive Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2020: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Biggs JJ, Hughes MH, et al. *Occupational Asthma*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2020: Available at: www.mdguidelines.com. Accessed March 15, 2025.
- Hegmann KT, Bruns D, Warren PA, et al. *Anxiety Disorders*. American College of Occupational

- and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2021: Available at: www.mdguidelines.com. Accessed March 15, 2025.
24. Hegmann KT, Weiss MS, Bowden K, et al. *Opioids*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2023: Available at: www.mdguidelines.com. Accessed March 15, 2025.
25. Hegmann KT, Melhorn JM, Ausfahl J, et al. *Hand, Wrist, and Forearm Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2023: Available at: www.mdguidelines.com. Accessed March 15, 2025.
26. Hegmann KT, Cowl CT, Harber P, et al. *COVID-19 (Coronavirus)*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. 2023. Reed Group, LLC; 2018: Available at: www.mdguidelines.com. Accessed March 15, 2025.
27. Hegmann KT, Hughes MH, Kaufman LR, et al. *Shoulder Disorders*. American College of Occupational and Environmental Medicine (ACOEM) Practice Guidelines. Reed Group, LLC; 2023: Available at: www.mdguidelines.com. Accessed March 15, 2025.
28. Harris JS, Weiss MS, Haas NS, et al. Methodology for ACOEM's occupational medicine practice guidelines—2017 revision. *J Occup Environ Med* 2017;59:913–919.
29. American College of Occupational and Environmental Medicine. Methodology. 2025. Available at: <https://acoem.org/acoem/media/PDF-Library/ACOEM-Methodology-Aug-23-2025-Final.pdf>. Accessed September 1, 2025.
30. Chou R, Loeser JD, Owens DK, et al. Interventional therapies, surgery, and interdisciplinary rehabilitation for low back pain: an evidence-based clinical practice guideline from the American Pain Society. *Spine (Phila Pa 1976)* 2009;1066–1077.
31. GRADE Working Group. Criteria for applying or using GRADE. 2016. Available at: https://www.gradeworkinggroup.org/docs/Criteria_for_using_GRADE_2016-04-05.pdf. Accessed May 15, 2025.
32. Brouwers MC, Kho ME, Browman GP, et al. AGREE II: advancing guideline development, reporting and evaluation in health care. *CMAJ* 2010;182:E839–E842.
33. Institute of Medicine (US) Committee on Standards for Developing Trustworthy Clinical Practice Guidelines, Graham R, Mancher M, Miller Wolman D, Greenfield S, Steinberg E, eds. *In: Clinical Practice Guidelines We Can Trust*. Washington, DC: National Academies Press; 2011.
34. Shea BJ, Grimshaw JM, Wells GA, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. *BMC Med Res Methodol* 2007;7:10.
35. Hegmann KT, Travis R, Belcourt RM, et al. Diagnostic tests for low back disorders. *J Occup Environ Med* 2019;61:e155–e168.
36. Burton AK, Waddell G, Tillotson KM, Summerton N. Information and advice to patients with back pain can have a positive effect. A randomized controlled trial of a novel educational booklet in primary care. *Spine (Phila Pa 1976)* 1999; 2484–2491.
37. Van Oosterwijck J, Nijs J, Meeus M, et al. Pain neurophysiology education improves cognitions, pain thresholds, and movement performance in people with chronic whiplash: a pilot study. *J Rehabil Res Dev* 2011;48:43–58.
38. Randhawa K, Côté P, Gross DP, et al. The effectiveness of structured patient education for the management of musculoskeletal disorders and injuries of the extremities: a systematic review by the Ontario Protocol for Traffic Injury Management (OPTIMA) Collaboration. *J Can Chiropr Assoc* 2015;59:349.
39. Yu H, Côté P, Southerst D, et al. Does structured patient education improve the recovery and clinical outcomes of patients with neck pain? A systematic review from the Ontario Protocol for Traffic Injury Management (OPTIMA) Collaboration. *Spine J* 2016;16:1524–1540.
40. Waddell G, Burton AK. Occupational health guidelines for the management of low back pain at work: evidence review. *Occup Med (Lond)* 2001;51:124–135.
41. Anema JR, Steenstra IA, Bongers PM, et al. Multidisciplinary rehabilitation for subacute low back pain: graded activity or workplace intervention or both? A randomized controlled trial. *Spine (Phila Pa 1976)* 2007;32:291.
42. Evjenth O, Hamberg J. *Muscle Stretching in Manual Therapy: A Clinical Manual*. Alfta, Sweden: Alfta Rehab; 1984.
43. Bernacki EJ, Yuspeh L, Tao X. Determinants of escalating costs in low risk workers' compensation claims. *J Occup Environ Med* 2007;49:780–790.
44. Hlobil H, Staal JB, Spoelstra M, Ariens GA, Smid T, van Mechelen W. Effectiveness of a return-to-work intervention for subacute low-back pain. *Scand J Work Environ Health* 2005;31:249–257.
45. Steenstra IA, Anema JR, van Tulder MW, Bongers PM, de Vet HC, van Mechelen W. Economic evaluation of a multi-stage return to work program for workers on sick-leave due to low back pain. *J Occup Rehabil* 2006;16:557–578.
46. Steenstra IA, Anema JR, Bongers PM, de Vet HC, van Mechelen W. Cost effectiveness of a multi-stage return to work program for workers on sick leave due to low back pain, design of a population based controlled trial [ISRCTN60233560]. *BMC Musculoskelet Disord* 2003;4:26.
47. Feuerstein M, Huang GD, Ortiz JM, Shaw WS, Miller VI, Wood PM. Integrated case management for work-related upper-extremity disorders: impact of patient satisfaction on health and work status. *J Occup Environ Med* 2003;45:803–812.
48. Wickizer TM, Franklin G, Fulton-Kehoe D, Turner JA, Mootz R, Smith-Weller T. Patient satisfaction, treatment experience, and disability outcomes in a population-based cohort of injured workers in Washington state: implications for quality improvement. *Health Serv Res* 2004;39: 727–748.
49. Owens JD, Hegmann KT, Thiese MS, Phillips AL. Impacts of adherence to evidence-based medicine guidelines for the management of acute low back pain on costs of worker's compensation claims. *J Occup Environ Med* 2019;61:445–452.
50. Gaspar FW, Thiese MS, Wizner K, Hegmann K. Guideline adherence and lost workdays for acute low back pain in the California workers' compensation system. *PLoS One* 2021;16: e0253268.
51. Gaspar FW, Kownacki R, Zaidel CS, Conlon CF, Hegmann KT. Reducing disability durations and medical costs for patients with a carpal tunnel release surgery through the use of opioid prescribing guidelines. *J Occup Environ Med* 2017;59:1180–1187.
52. Phillips AL, Thiese MS, Freeman M, Kartchner R, Hegmann KT. Implementation of an opioid guideline impacts on opioid prescriptions, adverse outcomes, and an association with a state opioid-related fatalities. *J Occup Environ Med* 2019;61: 653–658.
53. Washington State Department of Labor & Industries. *Attending Provider's Return to Work Desk Reference*. 2004. Available at: <https://d2l2jhoszs7d12.cloudfront.net/state/Washington/Washington%20State%20Department%20of%20Labor%20and%20Industries/200-002-000.pdf>. Accessed May 15, 2025.
54. Daley D, Payne LP, Galper J, et al. Clinical guidance to optimize work participation after injury or illness: the role of physical therapists. *J Orthop Sports Phys Ther* 2021;51:CPG1–CPG102.
55. Lee AC, Deutsch JE, Holdsworth L, et al. Telerehabilitation in physical therapist practice: a clinical practice guideline from the American Physical Therapy Association. *Phys Ther* 2024; 104:pzae045.
56. Hymel PA, Stave GM, Burton WN, et al. Incorporating lifestyle medicine into occupational medicine practice: ACOEM guidance statement. *J Occup Environ Med* 2025;67:e72–e84.
57. van Vilsteren M, van Oostrom SH, de Vet HC, Franche RL, Boot CR, Anema JR. Workplace interventions to prevent work disability in workers on sick leave. *Cochrane Database Syst Rev* 2015;2015:CD006955.
58. Hurwitz N. Predisposing factors in adverse reactions to drugs. *Br Med J* 1969;1:536–539.
59. Aljadhey H, Mahmoud MA, Mayet A, et al. Incidence of adverse drug events in an academic hospital: a prospective cohort study. *Int J Qual Health Care* 2013;25:648–655.
60. Johnell K, Klarin I. The relationship between number of drugs and potential drug-drug interactions in the elderly: a study of over 600,000 elderly patients From the Swedish prescribed drug register. *Drug Saf* 2007;30:911–918.
61. Mannes CK, Derckx FH, de Ridder MA, Man in 't Veld AJ, van der Cammen TJ. Contribution of adverse drug reactions to hospital admission of older patients. *Age Ageing* 2000;29:35–39.
62. Budnitz DS, Pollock DA, Weidenbach KN, Mendelsohn AB, Schroeder TJ, Anest JL. National surveillance of emergency department visits for outpatient adverse drug events. *JAMA* 2006;296:1858–1866.
63. Nolan L, O'Malley K. Prescribing for the elderly part I: sensitivity of the elderly to adverse drug reactions. *J Am Geriatr Soc* 1988;36:142–149.
64. Green JL, Hawley JN, Rask KJ. Is the number of prescribing physicians an independent risk factor for adverse drug events in an elderly outpatient population? *Am J Geriatr Pharmacother* 2007; 5:31–39.
65. Peters JR, Hixon DR, Conner DP, Davit BM, Catterson DM, Parise CM. Generic drugs—safe, effective, and affordable. *Dermatol Ther* 2009; 22:229–240.
66. Meredith PA. Generic drugs. *Drug Saf* 1996;15: 233–242.
67. Hegmann KT, Travis R, Andersson GBJ, et al. Non-invasive and minimally invasive management of low back disorders. *J Occup Environ Med* 2020; 62:e111–e138.
68. Tian R, Zheng F, Zhao W, et al. Prevalence and influencing factors of nonunion in patients with tibial fracture: systematic review and meta-analysis. *J Orthop Surg Res* 2020;15:377.
69. Bhattacharyya T, Levin R, Vrahas MS, Solomon DH. Nonsteroidal antiinflammatory drugs and nonunion of humeral shaft fractures. *Arthritis Rheum* 2005;53:364–367.
70. Jarupongprapa S, Ussavassodhi P, Katchamart W. Comparison of gastrointestinal adverse effects between cyclooxygenase-2 inhibitors and non-selective, non-steroidal anti-inflammatory drugs plus proton pump inhibitors: a systematic review and meta-analysis. *J Gastroenterol* 2013;48:830–838.
71. Nissen SE, Yeomans ND, Solomon DH, et al. Cardiovascular safety of celecoxib, naproxen, or ibuprofen for arthritis. *N Engl J Med* 2016;375: 2519–2529.
72. Scarpignato C, Lanis A, Blandizzi C, Lems WF, Hermann M, Hunt RH. Safe prescribing of

- non-steroidal anti-inflammatory drugs in patients with osteoarthritis—an expert consensus addressing benefits as well as gastrointestinal and cardiovascular risks. *BMC Med* 2015;13:55.
73. Aronson JK. Inhibiting the proton pump: mechanisms, benefits, harms, and questions. *BMC Med* 2016;14:1–4.
74. Busse JW, Sadeghirad B, Oparin Y, et al. Management of acute pain from non-low back, musculoskeletal injuries: a systematic review and network meta-analysis of randomized trials. *Ann Intern Med* 2020;173:730–738.
75. Busse JW, Wang L, Kamaleldin M, et al. Opioids for chronic noncancer pain: a systematic review and meta-analysis. *JAMA* 2018;320:2448–2460.
76. Kool J, Bachmann S, Oesch P, et al. Function-centered rehabilitation increases work days in patients with nonacute nonspecific low back pain: 1-year results from a randomized controlled trial. *Arch Phys Med Rehabil* 2007;88:1089–1094.
77. Mair F, Whitten P. Systematic review of studies of patient satisfaction With telemedicine. *Br Med J* 2000;320:1517–1520.
78. DelliFraine JL, Dansky KH. Home-based telehealth: a review and meta-analysis. *J Telemed Telecare* 2008;14:62–66.
79. Hersh W, Helfand M, Wallace J, et al. A systematic review of the efficacy of telemedicine for making diagnostic and management decisions. *J Telemed Telecare* 2002;8:197–209.
80. Hersh WR, Helfand M, Wallace J, et al. Clinical outcomes resulting from telemedicine interventions: a systematic review. *BMC Med Inform Decis Mak* 2001;1:5.
81. Hersh WR, Hickam DH, Severance SM, Dana TL, Krages KP, Helfand M. Telemedicine for the Medicare population: update. *Evid Rep Technol Assess (Full Rep)* 2006;131:1–41.
82. Hailey D, Roine R, Ohinmaa A. Systematic review of evidence for the benefits of telemedicine. *J Telemed Telecare* 2002;8(Suppl 1):1–30.
83. Whitten PS, Mair FS, Haycox A, May CR, Williams TL, Hellmich S. Systematic review of cost effectiveness studies of telemedicine interventions. *BMJ* 2002;324:1434–1437.